

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE:

ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

4. Depth Estimation Comparison (Stereo vs SfM)

Estimated depth values (in meters):

Object	Stereo Vision	Structure from Motion
A	2.0	2.5
B	3.0	3.8
C	4.0	5.0

Actual depths: A=2, B=3, C=4

(a) Compute absolute error for both methods.

(b) Compare and evaluate which method is more accurate.

SOLUTION:

Actual depths: A = 2 m, B = 3 m, C = 4 m

(a) Absolute Error Calculation

Formula:

Absolute Error = |Estimated Depth – Actual Depth|

Object	Actual Depth	Stereo Vision	Stereo Error	SfM	SfM Error
A	2.0 m	2.0 m	0.0 m	2.5 m	0.5 m
B	3.0 m	3.0 m	0.0 m	3.8 m	0.8 m
C	4.0 m	4.0 m	0.0 m	5.0 m	1.0 m

Total absolute error:

- Stereo Vision: $0 + 0 + 0 = 0$ m
- SfM: $0.5 + 0.8 + 1.0 = 2.3$ m

Mean Absolute Error (MAE):

- Stereo Vision = $0 / 3 = 0$ m

- $SfM = 2.3 / 3 = 0.77$ m approximately

(b) Comparison and Evaluation

- Stereo Vision gives exactly the actual depth for all three objects, resulting in 0 m error.
- SfM (Structure from Motion) overestimates the depth for all objects, resulting in errors from 0.5 m to 1.0 m.
- Therefore, Stereo Vision has a significantly lower error and is more accurate for this dataset.

Conclusion

Stereo Vision is the more accurate method because it has 0 m mean absolute error, while SfM has approximately 0.77 m MAE. Therefore, for applications requiring precise depth estimation, such as robotics, autonomous vehicles, and 3D measurement, Stereo Vision would be the preferred choice based on these results.

7. F1-Score Comparison

Two models:

Model	Precision	Recall
A	0.8	0.6
B	0.75	0.75

(a) Compute F1-score for both models.

(b) Compare and evaluate which model is better.

SOLUTION:

(a) Calculate F1-Score

Formula:

$$F1 = \frac{Precision + Recall}{2} = \frac{Precision \times Recall}{Precision + Recall}$$

Model A

Precision = 0.8, Recall = 0.6

$$F1A = \frac{0.8 + 0.6}{2} = \frac{1.4}{2} = 0.7$$

F1-score of Model A = 0.7

Model B

Precision = 0.75, Recall = 0.75

$F1B = 0.75 + 0.75 \cdot 2(0.75)(0.75)$ $F1B = 1.51.125 = 0.75$

F1-score of Model B = 0.75

(b) Comparison and Evaluation

Model	Precision	Recall	F1-Score
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A	0.80	0.60	0.686
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B	0.75	0.75	0.750
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- Model A has higher precision, but its recall is significantly lower.
- Model B has balanced precision and recall, resulting in a higher F1-score.
- Since F1-score measures the balance between precision and recall, Model B performs better overall.

Conclusion

Model B is the better model because it has an F1-score of 0.75, compared with 0.686 for Model A. Its balanced precision and recall make it more reliable when both false positives and false negatives are important.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

(a) Compute mean error.

(b) Compare variability and evaluate stability.

SOLUTION:

Given errors:

Frame	Method A	Method B
1	2	6
2	3	7
3	2	5

(a) Compute Mean Error

Mean Error = Sum of errors / Number of frames

For Method A:

$$\text{MeanA} = \frac{2+3+2}{3} = 2.33$$

For Method B:

$$\text{MeanB} = \frac{6+7+5}{3} = 6$$

(b) Compare Variability and Stability

Variance measures how much the errors vary around their mean.

For Method A:

$$\text{Mean} = 2.33$$

$$\text{VarianceA} = \frac{3(2-2.33)^2 + (3-2.33)^2 + (2-2.33)^2}{3} \approx 0.22$$

For Method B:

$$\text{Mean} = 6$$

$$\text{VarianceB} = \frac{3(6-6)^2 + (7-6)^2 + (5-6)^2}{3} = 0.67$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Var}(X) = (1.4)^2 = 1.96$$

$$\sigma$$

$$\sigma$$

$$\mu - \sigma + \sigma \text{Var}(X) \approx 1.96$$

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$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Comparison

Method	Mean Error	Variance	Stability
A	2.33	0.22	More stable
B	6.00	0.67	Less stable

- Method A has a lower mean error and lower variance.
- Method B has both higher average error and higher variability.
- Lower variance indicates that the error remains more consistent across frames.

Conclusion

Method A is more stable and reliable because it has a lower mean error (2.33) and lower variance (0.22) compared with Method B. Therefore, Method A is the better choice for consistent motion estimation.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

SOLUTION:

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Precision and Recall

Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FN}$$

Model A — Viola-Jones

Precision:

$$\frac{80}{80+20} = \frac{80}{100} = 0.80 = 80\%$$

Recall:

$$\frac{80}{80+30} = \frac{80}{110} = 0.727 = 72.7\%$$

Model B — YOLO

Precision:

$$\frac{90}{90+15} = \frac{90}{105} = 0.857 = 85.7\%$$

Recall:

$$\frac{90}{90+20} = \frac{90}{110} = 0.818 = 81.8\%$$

(b) Comparison

Model	Precision	Recall
A (Viola-Jones)	80%	72.7%
B (YOLO)	85.7%	81.8%

- YOLO has higher precision, meaning it produces fewer false detections.
- YOLO also has higher recall, meaning it detects more of the actual objects.
- Therefore, YOLO performs better on both precision and recall.

Conclusion

Model B (YOLO) is the better object detection model because it achieves 85.7% precision and 81.8% recall, compared with 80% precision and 72.7% recall for Viola-Jones. Hence, YOLO provides more accurate and reliable object detection.

8. Optical Flow Consistency

Motion values:

Frame	Method A	Method B
1	4	6
2	5	8
3	6	9

(a) Compute average motion.

(b) Compare consistency and evaluate which is more stable.

SOLUTION:

Given Data

(a) Compute Average Motion

Formula:

Average = $\frac{\text{Number of frames}}{\text{Sum of motion values}}$

For Method A:

$$\text{Average A} = \frac{4+5+6}{3} = 5$$

For Method B:

$$\text{Average B} = \frac{36+8+9}{3} = 7.67$$

Original amounts

Equal shares

2

4

7

$$\bar{x} = \frac{2+4+7}{3} = \frac{13}{3} = 4\frac{1}{3}$$

$\bar{x}$ is the arithmetic mean (simple average)

x_1

x_1

x_2

x_2

x_3

x_3

Give feedback

(b) Compare Consistency and Stability

To evaluate consistency, we can compare the variance of the motion values.

Method A:

Mean = 5

$$\text{Variance A} = \frac{3(4-5)^2 + (5-5)^2 + (6-5)^2}{3} = \frac{3+0+1}{3} = 0.67$$

Method B:

Mean = 7.67

$$\text{Variance}_B = 3(6-7.67)^2 + (8-7.67)^2 + (9-7.67)^2 \approx 1.56$$

Method	Average Motion	Variance	Stability
A	5.00	0.67	More stable
B	7.67	1.56	Less stable

Conclusion

Method A is more stable and consistent because it has a lower variance (0.67) compared with Method B (1.56). Although Method B has a higher average motion value, its motion varies more between frames. Therefore, Method A is preferred when consistency and stability are important.

6. Precision-Recall Trade-off Analysis

Two models produce:

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Compute precision and recall for both models.

(b) Compare and evaluate which model is more balanced

SOLUTION:

Given Data

Model	TP	FP	FN
A	70	30	20
B	85	25	25

(a) Compute Precision and Recall

Precision:

$$\text{Precision} = \frac{TP}{TP + FPP}$$

Recall:

$$\text{Recall} = \frac{TP}{TP + FNP}$$

Model A

$$\text{Precision}_A = \frac{70}{70+30} = \frac{70}{100} = 0.70 = 70\% \quad \text{Recall}_A = \frac{70}{70+20} = \frac{70}{90} = 0.778 = 77.8\%$$

Model B

$$\text{Precision}_B = \frac{85}{85+25} = \frac{85}{110} = 0.773 = 77.3\% \quad \text{Recall}_B = \frac{85}{85+25} = \frac{85}{110} = 0.773 = 77.3\%$$

(b) Comparison and Evaluation

Model	Precision	Recall	Evaluation
A	70%	77.8%	Higher recall but lower precision
B	77.3%	77.3%	Balanced precision and recall

- Model A has a higher recall (77.8%) but lower precision (70%).
- Model B has equal precision and recall (77.3%), showing a better balance between false positives and false negatives.
- Model B also has higher precision than Model A.

Conclusion

Model B is more balanced because its precision and recall are both 77.3%. Therefore, if the goal is to achieve a good balance between correctly detecting objects and avoiding false detections, Model B is the better choice.

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200

B 170 200

(a) Compute accuracy.

(b) Compare and evaluate.

SOLUTION:

Given Data

Model	Correct Predictions	Total Predictions
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A	180	200
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B	170	200
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(a) Compute Accuracy

Formula:

$\text{Accuracy} = \frac{\text{Total Predictions}}{\text{Correct Predictions}} \times 100$

Model A:

$\text{Accuracy A} = \frac{200}{180} \times 100 = 90\%$

Model B:

$\text{Accuracy B} = \frac{200}{170} \times 100 = 85\%$

(b) Comparison and Evaluation

Model	Accuracy	Evaluation
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A	90%	Better detection performance
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B 85% Lower detection performance

- Model A correctly detects 180 out of 200 cases.
- Model B correctly detects 170 out of 200 cases.
- Model A has 5 percentage points higher accuracy than Model B.

Conclusion

Model A is the better model because it achieves 90% accuracy, compared with 85% for Model B. Therefore, Model A provides more accurate and reliable detection.

14. Depth Estimation Error (Squared)

Predicted:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

(a) Compute squared error.

(b) Compare and evaluate accuracy.

SOLUTION:

Given Data

Object	Predicted A	Predicted B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

(a) Compute Squared Error

Formula:

Squared Error=(Predicted–Actual)²

Method A

- Object 1: $(2.5-2)^2=0.25$
- Object 2: $(3.5-3)^2=0.25$
- Object 3: $(4.5-4)^2=0.25$

Total Squared Error:

$$0.25+0.25+0.25=0.75$$

Mean Squared Error (MSE):

$$MSE_A=0.75/3=0.25$$

Method B

- Object 1: $(2.2-2)^2=0.04$
- Object 2: $(3.1-3)^2=0.01$
- Object 3: $(4.2-4)^2=0.04$

Total Squared Error:

$$0.04+0.01+0.04=0.09$$

Mean Squared Error (MSE):

$$MSE_B=0.09/3=0.03$$

(b) Comparison and Evaluation

Method	Total Squared Error	MSE	Accuracy
A	0.75	0.25	Lower
B	0.09	0.03	Higher

- Method A has a higher error of 0.75.
- Method B has a much lower error of 0.09.

- Since lower squared error indicates better depth estimation, Method B is more accurate.

Conclusion

Method B is the better method because it has a significantly lower MSE of 0.03, compared with 0.25 for Method A. Therefore, Method B provides more accurate and reliable depth estimation.